

Review Article

Dengue Virus Infection: Immune Response and Therapeutic Targets

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Abstract. Flavivirus infection, especially dengue virus infection caused by DENV, is known to be a significant health concern globally owing to the high incidence and mortality rate. The expanding and increasing disease burden calls for the need to develop an effective treatment and prevent the event of fatal complications, including dengue hemorrhagic fever/dengue shock syndrome. The DENV-induced immune response has been described as paradoxical because it has a protective role in viral clearance but, at the same time, causes more severe infection through viral-specific immunity. This is further complicated by high homology and cross-reactivity between different serotypes of DENV, causing a more severe disease presentation during secondary infection by a heterologous serotype. This serotype complexity poses a challenge for the development of a universal flavivirus vaccine. This review highlights the significance of high motility group box 1 (HMGB1) and nucleotide-binding oligomerization domain, leucine rich repeat and pyrin domain containing 3 (NLRP3) inflammasome activation pathways in initiating an inflammatory response through the downstream activation of nuclear factor κ B and proinflammatory cytokine Interleukin (IL)-1B, IL-18 release in DENV infection. It also discusses the role of NLRP3 in activating cellular apoptosis and pyroptosis leading to systemic failure, especially in peripheral tissues. Over the decades, there has been much progress in understanding the immunopathogenesis of DENV infection. Researchers have been studying key pathogenic molecules for potential therapeutic targets including HMGB1 and NLRP3 inflammasome inhibitors, which is explored in this review. Ultimately, although there is not yet an effective antiviral or vaccine for DENV, immunomodulators continue to pave the way to decrease disease severity in infected individuals.

INTRODUCTION

Dengue virus (DENV) is the most dominant arthropod-borne infection worldwide, with a high incidence of global mortality and morbidity. It affects humans living in the tropical and sub-tropical countries of the world and is transmitted chiefly by the *Aedes aegypti* and *Aedes albopictus* mosquitos. Globally, an estimated 390 million individuals are infected annually, and 2.5 billion more are at risk of infection. Around 10 million individuals present with clinical symptoms, of whom 500,000 have severe presentations.¹ It causes severe illness ranging from dengue fever to dengue hemorrhagic fever (DHF) and life-threatening dengue shock syndrome (DSS).²

Dengue virus is a positive-sense, single-stranded RNA virus with a genome size of ~50 nm and is spherical in shape.¹ The genome consists of a long open reading frame that encodes a polypeptide chain. During replication, this polyprotein is further cleaved by proteases to form three structural proteins, namely capsid protein (C protein), envelope protein (E protein), and pre-membrane protein (prM protein), and seven nonstructural (NS) proteins, NS1, NS2A, NS2B, NS3, NS4A, NS4B, and NS5.² These proteins constitute the main structure of the virus and play a vital role in viral replication and pathogenesis of flavivirus infection. Dengue virus infection is initiated by the binding of the E protein of the virus to host cell receptors via ligand-mediated attachment. The virion is then internalized by Clathrin-mediated endocytosis to form an endosome. The low pH in the endosome then promotes membrane fusion and uncoating of the viral nucleocapsid into structural and non-structural proteins. Translation, replication, and assembly of the viral genome then takes place in the endoplasmic

reticulum (ER), where more copies of the viral genome are synthesized and packed into C proteins. These immature nucleocapsids then bud off from the ER to the Golgi apparatus through the secretory pathway, where they undergo maturation and N-linked glycosylation of prM and E proteins. Furin cleavage of prM takes place before the viral particle is released via exocytosis. Upon exiting, further maturation occurs, where changes are made to the conformation and size of the virion.¹

There are four antigenically distinct serotypes of DENV, namely DENV-1, DENV-2, DENV-3, and DENV-4. Interestingly, DENV exhibits a phenomenon known as antibody-dependent enhancement (ADE) mediated by the Fc receptor (FcR) on macrophages. This leads to a more severe presentation with higher risk of DHF/DSS in individuals infected by a different viral serotype from the primary infection. Antibodies generated from the primary infection allow homotypic immunity but enhance heterotypic immunity through the production of cytokines, enhanced viremia, and antigenemia.³

To date, it remains a challenge to develop a vaccine that is equally effective against all four DENV serotypes. In 2015, Dengvaxia (CYD-TDV) by Sanofi Pasteur (Bridgewater, NJ) was licensed for use for previously infected individuals between 9 and 45 years of age in endemic countries. Since then, the Food and Drug Administration and other regulatory boards have authorized its use in 20 other countries.^{4,5} It is a live, attenuated, tetravalent vaccine that consists of prM and E proteins of all four types of DENV incorporated into the yellow fever virus (YFV) 17D vaccine strain backbone. Results from phase 3 clinical trials revealed an efficacy rate of 66%. However, in contradiction to results from earlier studies, serotype-specific efficacy for DENV-2 was poor, with less than 43% efficacy, possibly due to a less-effective T-cell immune response against NS proteins from the YFV strain. A study among children younger than 9 years showed severe symptoms requiring hospitalization, suggesting that Dengvaxia mimics a primary infection in these children and causes more

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severe symptoms owing to ADE upon natural infection. Other vaccines that are currently under clinical trial include live, recombinant, tetravalent vaccines such as TV003 and DEN-Vax, inactivated TDENV-PIV vaccine, and recombinant protein and RNA vaccines such as DEN-80E and V180.⁴

To develop effective therapies for DENV infection, it is crucial to understand the complexity of DENV-induced host immune responses and the host factors involved. The immune response to DENV infection is described as paradoxical because it has a protective role in viral clearance but at the same time causes more severe infection through viral-specific immunity. This is further complicated by high homology and cross-reactivity.⁶ Therefore, it is important to achieve a balanced defense system that effectively reduces viral infection while minimizing disease severity and tissue and organ damage. This review provides a broad overview of the host immune response against DENV infections, challenges in developing dengue vaccines, the role of both viral and host factors, and the potential of drug repurposing.

Immune response to DENV infections.

Innate immunity and DENV immunogenicity. When DENV infects through a mosquito bite, immune sentinel cells in the epidermis, such as Langerhans cells, dendritic cells (DCs), and mast cells are the first to respond. Mast cells rapidly degranulate, releasing cytokines, chemokines, and proteases. These chemokines recruit cytotoxic cells, including CD8⁺ T cells, natural killer (NK) cells, and NK T cells to the infection site. Infected DCs, macrophages, monocytes, and keratinocytes produce additional cytokines.⁷ Dendritic cells also secrete MMP-2 and MMP-9 that facilitate migration to the lymph nodes and blood circulation.⁸ Dendritic cells express pattern recognition particles (PRRs) such as TLRs and C-type lectin receptors, leading to innate response activation and downstream release of interferons (IFNs) and tumor necrosis factor (TNF). Toll-like receptor 4 (TLR4) and TLR7 binding leads to the release of IFN- α/β , which activates the janus kinase/signal transducers and activators of transcription (JAK/STAT) pathway.⁸ Through IFN-receptor binding, autophosphorylation of tyrosine kinase Jak1 and Tyk2 takes place and leads to activation of transcription factors STAT1 and STAT2.⁹ This is a crucial step to activate effector protein production through autocrine and paracrine signaling. The STAT signaling pathway can be partially blocked by NS2A, NS4A, and NS4B.¹⁰ NS5 inhibits STAT2 transcription factor to suppress expression of IFN-stimulated genes (ISGs).⁸

Interferon- β production is also mediated by the melanoma differentiation-associated protein 5 (MDA5) and retinoic acid-inducible gene 1 (RIG-1) RNA helicases. Once viral RNA is recognized and binds to caspase activation and recruitment domains (CARDs), it is translocated to the mitochondria to bind with adaptor protein mitochondrial antiviral-signaling protein (MAVS).⁹ Here, RIG-1/MAVS signals the activation of IFN regulatory factor 3 (IRF3) and nuclear factor κ B (NF- κ B), which will then induce type I IFN (IFN-I) production in the nucleus. During secondary infection, non-neutralizing antibodies that are complexed with DENV inhibit the MDA5 and RIG-1 pathway and subsequent IFN-mediated responses.⁸ This response may be hijacked by DENV through immunogenicity of several viral proteins. NS4A effectively inhibits RIG-1/MAVS interaction through MAVS CARD domain binding. NS3 is capable of inhibiting RIG-1 translocation from the

cytosol to the mitochondria by having protein motif RxEP.^{8,9} NS2A and NS4B antagonize the phosphorylation of IRF3 and subsequent downstream responses, whereas protease NS2B/3 cleaves IRF3 activator and inhibits IFN- α/β production.

Despite DENV being an RNA virus, studies have also found the involvement of a DNA-sensing pathway in DENV infection. The DNA triggers the cyclic GMP-AMP synthase (cGAS)-stimulator of IFN genes (STING) signaling and downstream activation of NF- κ B and proinflammatory cytokine expression.⁹ Dengue virus NS2B3 is able to cleave STING and prevent RIG-1/MAVS response. NS2B on the other hand degrades cGAS to prevent IFN expression.⁹ Another mechanism that has been studied is evasion of the interference RNA (RNAi) pathway. This can be through either sub-genomic flavivirus RNA production to inhibit RNA cleavage or the expression of sub-structural NS4B, which modulates viral replication via the RNAi/micro-RNA pathway.⁸

In addition, there are cytokines that play a pathologic role and facilitate viral replication and dissemination, namely TNF- α and macrophage migration inhibitory factor (MIF). For example, there is a positive correlation between the levels of TNF- α and MIF and disease severity in patients presenting with severe dengue. Tumor necrosis factor- α has an additional role of exacerbating plasma leakage, increasing vascular permeability and coagulopathy.¹⁰ During the febrile early stages of infection, type 1 IFN release leads to viremia and elevated levels of NS1. The NS1 protein further enhances vascular permeability through MIF and TLR4 activation. Other cells that play a role include macrophages, mast cells, NK cells, and NK-like cells. Failure of early response to viral replication leads to elevated NS1 and viremia, amplification of cytokine release, and severe dengue.¹¹ These pathways highlight the close interplay between host innate immunity and the DENV structural proteins.

High motility group box 1 and NLRP3 inflammasome activation pathway. High motility group box 1 (HMGB1) has been recognized as one of the key modulators to initiate an inflammatory response through innate immunity and partially through adaptive immunity. In the nucleus, HMGB1 plays a role in transcription, replication, and recombination and promotes genomic stability through DNA interaction. However, once released into the extracellular environment through apoptosis or necrosis, it acts as a damage-associated molecular patterns (DAMP) to initiate host immune response. Macrophages and monocytes can also be stimulated by pathogen-associated molecular patterns (PAMPs) to release HMGB1.¹²

High motility group box 1-mediated inflammation occurs when extracellular HMGB1-DAMP/PAMP complexes bind to Receptor for Advanced Glycation Endproducts (RAGE) molecules, which are abundantly expressed in the lungs and are endocytosed into endosomes. Within these TLR4-containing endosomes, HMGB1 translocates to lysosomes to form endolysosomes that disrupt the membrane, allowing access to the cytosol.¹³ Another pathway involves the activation of TLR4 via the binding of disulfide HMGB1 with MD-2 receptors.¹⁴ These steps generate downstream proinflammatory release by forming a TLR4 complex for signal transduction. When secreted at low levels during initial stages of sepsis, HMGB1 has proinflammatory properties. However, at greater levels, this complex causes dysregulation of the

immune response manifested as immunosuppression and macrophage pyroptosis.¹⁵ Immunohistochemistry studies have revealed higher expression of cytoplasmic HMGB1 in postmortem peripheral tissue samples of the lung, liver, and heart in those with severe dengue, highlighting *in situ* participation of HMGB1 in severe dengue infection.¹² In macrophages and endothelial cells, HMGB1 signaling creates a positive feedback loop that leads to severe disease.

High motility group box 1 activates the nucleotide-binding oligomerization domain, leucine rich repeat and pyrin domain containing 3 (NLRP3) inflammasome pathway through direct downstream signaling of NF- κ B and activation of TLR2 and TLR4. Structurally, the NLRP3 inflammasome comprises NLRP3 (PYD-NOD-LRRs), apoptosis-associated speck-like protein containing a CARD (ASC) adaptor protein, and pro-caspase-1. The first step for NLRP3 action involves priming by cytosolic PRRs (i.e., TLR, NLR, RIG-1, interferon- α /beta receptor, and MDA5) and subsequent activation of NF- κ B and upregulation of pro-caspase-1, pro-interleukin (IL)-1 β , and pro-IL-18. This is followed by the activation of NLRP3 triggered by infection, damage to tissue, metabolic imbalances, and other stress signals. Once activated, pro-caspase-1 undergoes autocleavage to active caspase-1, which goes on to activate pro-IL-1 β and pro-IL-18 through proteolysis. These cytokines induce pyroptosis and the release of IL-1 β and IL-18 into the extracellular environment through the oligomerization of gasdermin D.^{6,16} Pan et al.¹⁷ demonstrated the importance of DENV M protein in NLRP3 activation, IL-1 β secretion, and increased vascular permeability in mice.

Generated IL-1 β , IL-18, and IL-23 then promote proinflammatory cytokine release by Th17/ $\gamma\delta$ T cells, therefore initiating host adaptive. IL-1 β is also an endogenous pyrogen that causes fever and systemic clinical presentation. In platelets, DENV infection activates the release of extracellular vesicle, which causes downstream release of neutrophil extracellular traps formations and increases in vascular permeability. Dengue virus NS1 protein amplifies the inflammasome response by sending autocrine TLR4 signals on platelets.⁶ High motility group box 1, together with other vasoactive cytokines and NS1, also acts cooperatively to further promote vascular permeability.⁶

The amplification of cytokine production leading to cytokine storm has been established as the main contributing factor for DHF/DSS. Heightened levels of cytokines, including TNF- α , IL-6, IL-8, IL-10, chemokine ligand (CCL)-2, CCL-3, chemokine (C-X-C motif) ligand (CXCL)-8, CXCL-10, and IFN- γ , leads to endothelial cell dysfunction and vascular permeability, causing hemorrhagic manifestations, shock, and death. These uncontrolled immune cell responses also aggravate hepatic dysfunction, hypotension, and thrombocytopenia.¹⁰

Adaptive immunity pathway. Humoral and cellular adaptive immunity are both essential for viral elimination. Observations from multiple studies have shown that there are similar yet distinct DENV serotypes with cross-reactivity. During a secondary heterologous infection, cross-reactive T cells specific to the primary DENV serotype become predominant. This expansion of preexisting cross-reactive and low-affinity memory T cells results in ineffective viral control and contributes to immunopathology and severe dengue disease.^{7,18} Results from preclinical studies on C57BL/6 mice showed

elevated liver enzymes and prolonged bleeding time, which support these findings.^{19,20}

After primary DENV infection, activated naive CD4⁺ and CD8⁺ T cells differentiate into effector T cells to eliminate viral infection through a direct or cytokine-mediated response. Cytotoxic CD8 and CD4 cells secrete granzyme and perforin B against major histocompatibility complex 1 (MHC1) or MHCII expressing DENV-infected cells.¹¹ CD8 T cells have epitope immunodominance against NS3, C protein, NS5, and NS4A/B proteins, whereas CD4 T cells have a preference for C protein followed by E, NS3, NS4A/B, and NS1.²⁰ Epitope recognition determines the amplitude of T-cell activation and disease severity. Downstream signaling of Th1 cells then produces IFN- γ and TNF- α , which have antiviral properties. Regulatory CD4T (Treg and Tr1 cells), on the other hand, release IL-10 and transforming growth factor- β to suppress immunity.²⁰ *In vivo* studies demonstrated the importance of T-cell response during acute infection seen via elevated levels of T-cell expression and activation markers (i.e., CD69).¹¹

Meanwhile, the B cells bind to DENV antigens, producing a primary IgM response followed by a more complex and specific IgG response. During secondary infections, the IgM response decreases, whereas the IgG response becomes more pronounced.²¹ These antibodies primarily target the NS1, E, and prM proteins of DENV.^{22,23} They aid in neutralization, complement fixation, antibody-dependent cell-mediated cytotoxicity, and ADE.^{11,19} Antibody-dependent enhancement leads to severe dengue via IFN- α blockade and increased IL-10, which lead to inhibition of immune response and increased viral replication, respectively. B-cell activation also leads to release of other protective cytokines such as IL-6, IFN- γ , and TNF- α and inhibitory IL-10 and IL-35.¹¹ IL-10 has been identified as the key cytokine that is significantly increased during acute dengue infection and is associated with progression to severe DHF. NS1 also induces IL-10 production by monocytes and is confirmed by the persistence of NS1 antigenemia and IL-10 during the initial febrile phase.²⁴

Challenges on the development of a dengue vaccine.

Antibody-dependent enhancement. After primary DENV infection, antibodies take up to 1 week to develop. Serotype-specific antibodies against the primary strain may persist for more than 60 years, but there is only temporary protection against other serotypes.²⁵ The preexisting neutralizing antibodies can block DENV from attaching to its natural receptor on the cell surface, thereby inhibiting virus entry. However, antibodies from a heterologous infection can be cross-reactive and facilitate ADE. During secondary infection, the DENV and non-neutralizing antibody complex bind to the FcR on macrophages and monocytes. This mechanism allows the virus to enter and escape the endosome in a manner similar to the primary infection pathway, resulting in a higher viral load and ultimately exacerbating the disease.

Another proposed mechanism of ADE activation is via the classical complement pathway that is seen in other viral infections such as HIV and Ebola. Antibody-dependent enhancement causes severe disease manifestations via downregulation of IL-12, TNF- α , IFN- γ , and nitric oxide radical secretion and at the same time enhancing IL-6 and IL-10 expression.⁸ IL-6 and IL-10 lead to the overexpression of suppressor of cytokine signaling-3 (SOC3), which inhibits the JAK/STAT pathway and induces a skewed Th2-type immune response that aids in viral

entry and produces sub-neutralizing antibodies.²⁶ Studies reveal that DENV entry is enhanced by both anti-E and anti-prM antibodies via Fc γ -mediated ADE. However, antibodies targeting the DENV envelope domain III (EDIII) induce a higher specificity and, more importantly, a lower risk of ADE.²⁷ After this, some researchers are steering away from the use of anti-prM and toward EDIII for the development of vaccines. For example, a tetravalent viruslike particle vaccine developed to display EDIIIs of all four DENV serotypes induced strong and durable multi-serotype neutralizing antibodies with decreased ADE in mice and macaques.²⁸

Cross-reactive antibody responses. Cross-reactivity of DENV with other flaviviruses also poses a challenge for vaccine development. The viral E protein is widely conserved among all flaviviruses and is the most exposed target for antibody development and neutralization. For instance, there is more than 50% homology noted between DENV and Zika virus (ZIKV), contributing to cross-reactivity.²⁹ This became evident during the 2016 ZIKV outbreak in DENV-endemic regions including Mexico and Brazil.^{30,31} The ADE of dengue disease is associated with low to intermediate titers of preexisting dengue antibodies, whereas high antibody titers protect against severe disease.^{22,32,33} Therefore, sub-neutralizing antibody levels together with cross-reactive memory B cells are the main immunopathological factors causing severe disease.

Immunomodulators.

Viral proteins. Envelope domain III has been hypothesized to be a potent virulence factor by inducing NLRP3 inflammasome-dependent vascular endothelial injury. This is supported by elevated EDIII levels in the plasma during peak viremia. Preclinical in vitro studies observed endothelial cell damage when EDIII was injected. Sequential EDIII and DENV-NS1 injection yielded further endothelial damage, liver dysfunction, hemorrhage, and other clinical symptoms of DHF.³⁴ Visterra (Cambridge, MA) engineered Ab513, a monoclonal antibody targeting EDIII domains of all four DENV serotypes. Preclinical studies have shown promising results with reduced viremia and DHF symptom severity even in the presence of cross-reactive antibodies. These antibodies were able to neutralize DENV-infected monocytes that express Fc γ receptors.^{35,36}

NS1 is a conserved glycoprotein associated with high virulence. It plays a role in promoting vascular permeability, coagulopathy, and thrombocytopenia, contributing to severe dengue. NS1-induced vascular leakage has been studied in recent years, with higher levels of soluble NS1 detected in severe dengue. There is a dose-dependent relationship between NS1 concentration and magnitude of vascular leakage due to endothelial dysfunction. Several possible mechanisms include via the activation of TLR4, MIF-mediated autophagy and junction disruption, and human platelet antigens-1-induced endothelial glycocalyx degradation.^{3,37} This is further exacerbated by cross-reaction of NS1 antibodies with endothelial cells to induce apoptosis.² However, several studies have yielded contradicting results, where low NS1 levels were detected despite severe clinical symptoms. This leaves a research gap that proposes the role of other cytokines in dengue pathogenesis. A study by Soe et al.³⁷ failed to note any correlation between NS1 levels and extent of vascular damage, but instead proposed that cytokines CCL2, CCL5, CCL20, and CXCL1 and intercellular adhesion molecule-1 also play a role in leukocyte infiltration and

disruption of inter-endothelial junctions. NN-DNJ, a potential drug target, is a derivative that binds to the conserved glycosylation sites N130 and N207 on NS1 to reduce release.² In addition, the use of monoclonal antibodies against NS1 is currently being explored.³

NS5 is the largest, most conserved protein and has an essential role in replication, RNA methylation, polymerization, and host immune system evasion. NS5 inhibits the JAK/STAT pathway by suppressing human signal transducer and activator of transcription 2 signaling and IFN production. The DENV NS5 protein binds to a STAT2 protein inhibiting phosphorylation, reduction of ISG transcription, and STAT2 degradation. During the cleavage of DENV polyprotein between NS5 and NS4B by protease NS2B/3, the N terminus of NS5 is exposed, and this is sufficient to cause degradation.^{38,39} Ashour et al.⁴⁰ also demonstrated that the ubiquitin-proteasome pathway is also involved through the action of the UBR box N-recogin-4 protein, resulting in STAT2 degradation. NS5 does not directly bind to STAT1 but blocks IFN- α -mediated STAT1 phosphorylation. X-ray crystallography and cryo-Electron Microscopy: revealed that NS5 is composed of an RNA methyltransferase (MTase) domain in the N-terminal, and an RNA-dependent RNA polymerase (RdRp) in the C-terminal that are linked together. This fusion domain makes it a unique and potential drug target.⁴¹ Unfortunately, the use of balapiravir, an RdRp inhibitor, in a study failed to show any significant difference in plasma viral load, cytokine profile, or fever clearance time when compared with a placebo group.⁴²

Recently, a highly potent pan-serotype DENV inhibitor, JNJ-A07, was discovered offering a possible breakthrough in DENV antiviral treatment. This molecule targets NS4B and inhibits de novo NS4B/NS3 complex formation but does not interfere with complexes that have been formed. In drug-treated infected mice models, there was an established reduction in RNA viral load in a dose-dependent manner and normalization of plasma proinflammatory cytokine levels (IL-18, IFN- γ , TNF, and IL-6). Even when the administration was delayed for several days, the antiviral potency was still maintained with a recorded drop in viral load. Other favorable pharmacokinetic properties include no adverse effects and very high barrier to resistance. Drug-resistant variants were only notable after more than 40 weeks of selection and at very low frequencies, making it unlikely for patients treated with this drug to develop resistance. Furthermore, the mutations in NS4B seemed to prevent the resistant variants from replicating in mosquito cells. This indicates that even if such mutants were to emerge, they might not be transmissible from human to human through the insect vector. To conclude, JNJ-A07 has very high potential, and further development of this compound should be pursued.⁴³

Host factors. Extracellular HMGB1 plays an essential role in the inflammatory pathway and can be targeted for treatment not only for infection but also for autoimmune diseases, cancer, and trauma. With TLR4 and RAGE identified as the main receptors, multiple strategies targeted against TLR4-HMGB1 or RAGE-HMGB1 complexes are being studied. Peptide P5779 inhibits HMGB1/TLR4-mediated inflammation by specifically binding to MD-2 and sparing the binding of the HMGB1/lipopolysaccharide (LPS) complex. This exclusively preserves endotoxin responsiveness, RAGE-mediated endocytosis, and downstream LPS-induced cytokine and chemokine

release.¹⁵ There is a concentration-dependent inhibition of MD2-HMGB1 interaction with sparing of PAMPs-mediated macrophage activation. The effective concentration was 5 μ g/mL noted by 50% suppression of TNF release.¹⁴ Studies conducted on global TLR4 gene-deficient mice revealed reduced arterial injury and upregulation of TLR4, HMGB1, and IL-6 expression, demonstrating that HMGB1 and TLR4 have a role in vascular inflammation.¹⁵ Recombinant HMGB1 box A protein has been used to treat preclinical models of inflammatory diseases. Although the mechanism of action remains unknown, studies on experimental inflammatory models such as common lymphoid progenitor (CLP) sepsis and arthritis have shown beneficial results. Anti-HMGB1 monoclonal antibodies (m2G7) works by binding to the box A epitope, affecting RAGE and TLR4 interactions and inhibition of TNF. In vivo studies on CLP sepsis in mice successfully proved improved survival when m2G7 was administered. Other positive findings included reduced memory impairments, brain pathology, and anemia post-acute CLP sepsis.¹⁵ Other small molecule targets for HMGB1 include ethyl pyruvate, glycyrrhizin, and aspirin that act by impeding HMGB1 function. Ethyl pyruvate inhibits HMGB1 secretion as seen in preclinical animal studies. Glycyrrhizin is derived from the licorice plant *Glycyrrhiza glabra* and inhibits extracellular HMGB1 action.⁴⁴ Treatment with glycyrrhizin conferred protection against inflammation induced by ischemia or reperfusion in liver and brain tissues in rat models.^{45,46} Aspirin is an irreversible COX 1 and 2 inhibitor and impedes the activation of disulfide HMGB1 and downstream inhibition of notably COX 2.⁴⁴

NLRP3 has a strong inflammatory response and hence has therapeutic potential. Some key targeted pathways include NLRP3 transcription and licensing inhibition via blockage of TLR and TNF-mediated response, NLRP3 activation inhibition via regulation of post-translational modification, and inhibition of downstream effector mechanisms. The drawback of all these approaches is the lack of specificity, which leads to many unintentional off-target effects. Targeting a common molecule is not ideal, as it would lead to inhibition of other non-NLRP3 inflammasome effectors, causing immune suppression.^{47,48} As of this date, there are three available biologics in the market against IL-1 β , namely anakinra (recombinant IL-1 receptor antagonist), canakinumab (IL-1 β antibody), and rilonacept (soluble decoy IL-1 receptor).

In targeting IL-1 β , some drawbacks need to be noted. First, inflammasome activation also leads to the release of other proinflammatory cytokines, including IL-18, that will not be targeted. In addition, the production of IL-1 β is nonspecific and is also released by the activation of other non-NLRP3 inflammasomes.^{47,48} Small molecule inhibitors of caspase-1, such as pro-drugs VX-740 and VX-765, are also under research. These drugs act by inhibiting caspase-1 action of pro-IL-1 β /18 cleavage. Phase 1 and 2 clinical studies yielded positive results showing anti-inflammatory properties and a good pharmacokinetic profile but were stopped because long-term use was associated with liver injury.⁴⁹ Lastly, there is CP456773, the most studied NLRP3 inflammasome antagonist. Coll et al.⁵⁰ discovered that this compound inhibits adenosine triphosphate-triggered NLRP3-mediated IL-1 β release at an inhibitory concentration (IC₅₀) of 8.1 nM in monocytes. Interestingly, it has highly specific properties and does not affect other TLR-mediated pathways including interferon-inducible

protein AIM2, NLRC4, or NLRP1 inflammasome activation.^{47,48} Another study on an influenza A virus-infected mouse model also successfully showed the use of CP456773 in preventing injury to lung tissue without impeding viral clearance.¹⁶

The initial host cell responses to flavivirus infection involve the secretion of type I IFNs, which consist of IFN- α and IFN- β .³⁹ Dengue virus has been shown to evade the innate immune response by blocking type-I IFNs.⁵¹ Suppression of IFN- α / β -stimulated genes has been observed in DHF/DSS patients.⁵² In an animal study on rhesus monkeys, the administration of IFN- α 2a was shown to reduce DENV viremia in primates, suggesting that this may be a therapeutic strategy.⁵³

The phosphoinositide 3 kinase (PI3K)/Akt strain transforming (AKT)/mammalian target of rapamycin (mTOR) pathway plays an essential role in cell survival, cell growth, and megakaryopoiesis. A DENV infection impairs this axis in megakaryocytes, leading to apoptosis, low platelet production, and subsequently hemorrhagic complication. A study on MEG-01-infected DENV cells confirmed this association, allowing it to be a potential antiviral target.⁵⁴ X-11-5-27, a synthetic compound works by inhibiting ASC binding to NLRP3 and promotes autophagy via the PI3K/AKT/mTOR pathway and prevents NLRP3 activation by inhibiting reactive oxygen species release.⁴⁹ In addition, niclosamide was previously found to have pharmacological efficacy against ZIKV. Kao et al.⁵⁵ conducted a study using in vivo and in vitro models and discovered that this drug inhibited DENV replication by inhibiting endosomal acidification and multiple cellular signaling pathways including mTOR, STAT3, and NF- κ B.

After DENV infection, the production of IL-10 leads to immune suppression and enhanced viral replication via activation of the IL-10/SOCS3 pathway. This may be exacerbated by Fc γ receptor-mediated ADE. As such, the use of neutralizing antibodies, inhibitors, or antagonists against IL-10 offers a viable therapeutic target, especially in severe dengue.^{56,57}

Drug repurposing. Drug repurposing is the identification of new therapeutic uses for drugs that are already in the market for different indications or compounds that failed to meet outcomes in clinical trials. Benefits include safer, cheaper, and less time-consuming medications given the increased likelihood of advancement to late-stage drug development. This approach has been widely studied for use against neglected tropical diseases primarily affecting developing countries, including DENV infection.⁵⁸ Numerous research studies have been conducted to evaluate the potential of repurposed drugs for the treatment of DENV infections, with several advancing to clinical trials (Supplemental Table 1).^{59–81} Although many preclinical studies have shown beneficial outcomes, there is still lack of evidence that these compounds have clinical superiority compared with placebo.

Targeting the vector. The spread of arthropod-borne disease is no longer confined to countries in tropical and subtropical regions owing to various factors including urbanization, climate change, movements of people and goods, and insecticide resistance. As a result, there is an increasing need for disease control to prevent further outbreaks.⁵ Over the decades, much progress has been achieved in limiting arboviral diseases; targeting the mosquito vector appears to be one of the most promising approaches. For dengue control, *Ae. aegypti* mosquitoes have been transinfected with the intracellular bacterium *Wolbachia*, an endosymbiotic

bacterium found naturally in over 60% of insect species but not in the *Ae. aegypti* mosquito.⁸² *Wolbachia*-infected mosquitoes limit transmission by inhibiting DENV replication through rapid viral RNA degradation mediated by the XRNI enzyme found in mosquitoes. In Indonesia, the ongoing AWED trial (Applying *Wolbachia* to Eliminate Dengue) is being studied to assess the efficacy of this intervention.¹ An randomized controlled trial conducted in Yogyakarta, Indonesia, comparing wMel *Wolbachia* strain-infected *Ae. aegypti* with other wild-type counterparts successfully showed lower rates of virologically confirmed dengue in the study group with a protective efficacy of 77.1% for all dengue serotypes.⁸³ *Wolbachia*-infected mosquitoes have also been released in endemic parts of Australia to control the spread of dengue. Similarly, in Niterói, Brazil, since the wMel deployments from 2017 to 2019, there have been 69%, 56%, and 37% reductions in dengue, chikungunya, and Zika incidence, respectively.⁸⁴

CONCLUSION

To conclude, DENV infections are a significant health concern, and more research should be conducted to find a solution to decrease the burden of this disease. As highlighted above, the host immune response plays a vital role in eradicating infection but at the same time contributes to disease severity. It is important to recognize the effector molecules and signaling pathways involved in the immunopathogenesis. The DENV has several mechanisms to evade host immune response and exacerbate endothelial dysfunction via the roles of structural and nonstructural molecules including NS1 and NS5. There is an established role of HMGB1 and NLRP3 inflammasome pathways in initiating inflammation. Research aimed at inhibiting these pathways offers a great potential to prevent downstream signaling and cytokine release. Efforts to develop an effective DENV vaccine has been hampered by the existence of four distinct serotypes that contribute to the mechanism of ADE and cross-reactivity. Future preclinical and clinical studies are therefore necessary to understand the immunopathogenesis and to optimistically discover more potential targets. Other efforts in therapeutic antiviral treatment focus on drug repurposing and targeting mosquito vectors.

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